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# Spin within systematic review abstracts on antiplatelet therapies after acute coronary syndrome: A cross-sectional study

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**Short Title:** Spin in systematic reviews regarding antiplatelet therapies

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## Abstract

**Objectives:** Spin is a reporting practice in which study results are misrepresented by overestimating efficacy or underestimating harm. Prevalence of spin varies between clinical specialties, and estimates are based almost entirely on clinical trials. Little is known about spin in systematic reviews.

**Design:** We performed a cross-sectional analysis searching MEDLINE and Embase for systematic reviews and meta-analyses pertaining to antiplatelet therapies following acute coronary syndrome. Data were extracted evaluating the presence of spin and study characteristics, including methodological quality as rated by AMSTAR-2. All data extraction was conducted in a masked, duplicate manner.

**Participants and Setting:** Not applicable.

Primary and Secondary Outcome Measures: We assessed abstracts of systematic reviews on antiplatelet therapy following acute coronary syndrome and evaluated the prevalence of the 9 most severe types of spin. We additionally explored associations between spin and certain study characteristics, including quality.

**Results:** Our searches returned 15,263 articles, and 185 systematic reviews met inclusion criteria. Of these 185 reviews, 31.9% (59/185) contained some form of spin in the abstract. Seven forms of spin (1, 2, 3, 4, 5, 7, and 9) among the 9 most severe were identified. No instances of types 6 or 8 were found. There were no statistically significant relationships between spin and the evaluated study characteristics or AMSTAR-2 appraisals.

**Conclusions:** Spin was present in abstracts for systematic reviews and meta-analyses; subsequent studies are needed to identify correlations between spin and specific study


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characteristics. Implementing changes will ensure that spin is reduced in the field of cardiology as well as other fields of medicine.

## Article Summary

### Strengths and Limitations of this Study:

- We strictly adhered to our protocol, which was published prior to data extraction on OSF[12]. We made our protocol, data, analysis scripts, data extraction forms, and other study artifacts available online to increase our transparency and reproducibility.
- As recommended by Cochrane Collaboration, we used an independent group to verify our results and implemented both screening and data extraction in a masked, duplicate manner to strengthen our study and minimize bias.
- The effect of spin on the reader's opinions and medical recommendations could not be judged, and the degree of spin use may have affected readers differently. To help mitigate the subjectivity of spin, we used online coursework and training to prepare our team for screening and data extraction.
- Our study had a cross-sectional design that did not allow for cause and effect analysis or generalizations to be made to other systematic reviews published in other journals.
- Selection bias may have negatively influenced the data collection. However, this effect was reduced due to the extensive training on methodology and data collection performed prior to extraction.

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**Competing Interest:**

None declared.

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## Introduction

As the gold standard in evidence-based medicine, regulating bodies often use systematic reviews to develop clinical practice guidelines that physicians refer to for treatment decisions [1,2]. Hence, the findings contained in systematic reviews and in their abstracts directly influence the care that patients receive and their prognosis.

The systematic reviews themselves serve as a means to collate all relevant primary data on a specific topic, critically appraise the literature, and condense the findings into a useful synthesis. A major issue for patient care is that the reviews often contain outcome reporting bias; that is, they share only positive findings, show selection bias, and/or transfer bias from primary data [3]. This bias is further complicated by clinicians having limited access to full-text articles, as well as limited time to read them in decision-making contexts. A 2013 study, for example, found that nearly 75% of physicians read the abstract alone, rather than the full-text of the article [4]. Importantly, abstracts have been found to not always accurately represent the findings reported in articles. For example, Toma and colleagues reported that 41% of the abstracts of randomized controlled trials contained different efficacy estimates than the full text [5]. Consistent with that report, other studies have shown that over 50% of abstracts had discrepancies in the results compared with the full text, and only 50% of abstracts discussed the side effects or harms experienced during clinical trials [6,7]. Considering the importance of evidence-based medicine, the necessity for reducing bias, especially in the abstracts of systematic reviews, cannot be overstated.

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One form of bias that is of increasing interest to systematic reviewers, clinical trial researchers, and epidemiologists is spin. Spin involves misrepresenting research results to mislead readers into favoring a certain outcome. It has been identified in multiple studies, particularly clinical trial research [8,9].

Accurate and objective research findings are necessary to guide patient care. For example, patients who have experienced acute coronary syndrome (ACS) with subsequent initiation of antiplatelet therapy are particularly at risk for morbidity and mortality arising from antiplatelet therapy inaccuracies. If clinicians misinterpret (or are misled by) clinical research on antiplatelet therapy, the results could be deadly.

However, no study has analyzed spin in this body of literature. Thus, the primary objective of this study was to analyze the presence of spin in the abstracts of systematic reviews on antiplatelet therapy following ACS. The secondary objective was to assess the study characteristics, including journal impact factor, funding source, and article type, associated with spin. We additionally describe methods for reducing spin in antiplatelet publications.

## Methods

### *Search Strategy*

A systematic review librarian (DW) constructed the search strategies for the MEDLINE (Ovid) and Embase (Ovid) databases to locate systematic reviews and meta-analyses pertaining to antiplatelet therapies (Figure 1). DW conducted these searches on June 2, 2020. Following the execution of these search strategies, the results were combined and uploaded into a screening

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platform, Rayyan (https://rayyan.qcri.org/). The duplicates were removed, and the titles and abstracts were independently screened for eligibility by 2 researchers (AW and DM) [10]. This process was performed in a masked, duplicate manner. Once the initial screening was complete, AW and DM resolved disagreements through discussion.

*Eligibility Criteria*

The following inclusion criteria had to be met for an article to be eligible for this study. The article must (1) be a systematic review with or without a meta-analysis; (2) focus on antiplatelet, anticoagulation, or thrombolytic therapies following ACS; (3) be conducted on human subjects only; (4) be available in English; and (5) have an abstract. The following study types were excluded from our study: observational studies (case-control, cohort, surveys), clinical trials, narrative reviews, and systematic reviews not related to antiplatelet therapies. We also excluded duplicates, withdrawn or retracted studies, letters to the editor, and any remaining article that did not meet the inclusion criteria. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA) definitions of systematic reviews and meta-analyses were used [11]. The selected studies were then uploaded to STATA for randomization, and as long as the previous criteria were met, the first 200 articles were sequentially extracted.

*Training*

Several levels of training were completed to ensure the quality of this study. Prior to the start of the study, all researchers were trained on an overview of study designs in clinical research, deduplication, and screening on Rayyan, and publication of protocols using Open Science Framework (OSF) [12]. The 2 researchers who conducted the screening (AW and DM) also

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completed an online training course on systematic reviews and meta-analyses offered by John Hopkins University on the Coursera platform [13]. Following the course, the researchers undertook 2 days of online and in-person training. The training session covered the definition and interpretation of the 9 most severe forms of spin in systematic review abstracts as defined by Yavchitz et al [14]. Finally, data extractors were trained on how to analyze a systematic review via A MeaSurement Tool to Assess systematic Reviews (AMSTAR-2). The AMSTAR tool was used to determine the methodological quality of each included systematic review and meta-analysis. A detailed outline of the training can be found in our study protocol.

*Identification of Spin and Data Extraction*

Data extraction was completed by AW and DM, using a pilot-tested Google form to evaluate for the 9 types of spin. The following general information was extracted from all reviews: the intervention type (surgery, pharmacologic, and nonpharmacologic); the year in which the review was received by the journal; the funding source (industry, private, public, none, not-mentioned, and hospital); whether the review discussed adherence to PRISMA [15] or PRISMA for abstracts [16]; whether the journal recommended adherence to PRISMA; and the journal's 5-year impact factor. The 9 types of spin are defined in Table 1. Any disagreements about the presence of spin were reconciled between AW and DM, and if an agreement could not be reached WA and MV acted as mediators. Following spin data extraction, the quality of each systematic review was analyzed using AMSTAR-2 (https://amstar.ca/) [17]. AMSTAR-2 has previously been used to evaluate systematic reviews in multiple medical disciplines, including surgery, cancer, psychiatry, pain management, and general medical interventions [17–21]. Based on scores from the 16-item scale, reviews were given confidence ratings of high, moderate, low, or critically low [17].

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Interrater reliability scores (assessed by kappa) for AMSTAR-2 ranged from .31 to 1.0 [17], with additional studies supporting its reliability, finding moderate interrater reliability (median K = 0.56) among multiple raters. Construct validity coefficients were high (AMSTAR, r = 0.91; Risk of Bias in Systematic Reviews [ROBIS], r = 0.84) [22]. All study characteristics, spin data, and AMSTAR-2 data were extracted in a masked, duplicate manner [14].

*Statistical Analysis*

The overall frequency of spin, study characteristics, and AMSTAR-2 ratings were analyzed using descriptive statistics, and the results are reported as both frequency counts and percentages. Following the screening, we included 185 systematic reviews. Binary logistic regression was used to determine the strength of associations via odds ratios between the occurrence of spin in a study's abstract with journal impact factor, study characteristics, and AMSTAR-2 appraisal rating, and a multivariable model was used to control for multiple characteristics. Prior to the start of the study, we determined that 185 articles were needed to power the analyses using GPower 3.1.9.7. Our analytic decisions are included in our protocol, and Stata 16.1 (StataCorp, LLC, College Station, TX) was used for all analyses.

*Patient and Public Involvement*

This study did not include any human participants or public involvement and therefore did not meet the regulatory definition of human subject research per the U.S. Code of Federal Regulations and was not subject to institutional review board oversight.

*Oversight, Transparency, Reproducibility, and Reporting*

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To ensure that our study is transparent and reproducible, we have made the protocol, extraction forms, data analysis scripts, and other study artifacts available via OSF. Our data and analysis scripts were sent to an independent investigation team and reanalyzed in a masked fashion to further the reproducibility. This study was conducted in tandem with other studies evaluating the presence of spin in systematic reviews in other medical fields. These studies adhered to a common methodology, which has been described elsewhere. While drafting this article, we integrated the relevant reporting guidelines from PRISMA [23] and Murad and Wang's [24] guidelines for meta-epidemiological studies.

## Results

*General Characteristics*

Our search queries returned 15,263 articles and 12,913 remained after deduplication. Of these, 12,372 were excluded after the initial title and abstract screening, which left 541 unique systematic reviews. A full-text review of the articles further excluded 356 reviews. Finally, 185 systematic reviews and meta-analyses pertaining to antiplatelet therapies following ACS remained and were included in data extraction. Our exclusion list from the screening process with rationale can be found in Figure 2.

All studies meeting inclusion criteria analyzed pharmacologic interventions. Of the 185 reviews, 61 (32.9%) systematic reviews were published in journals that recommended adherence to PRISMA, and 79 (42.7%) of the papers mentioned adherence to PRISMA or PRISMA-A. The majority of the systematic reviews and meta-analyses did not mention a funding source (93/185, 50.2%), followed by systematic reviews that did not receive any kind of funding (44/185, 23.8%). Of the papers that mentioned a funding source (92/185, 49.7%), the most common

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funding source was private funding (20/185, 10.8%), followed by public funding (17/185, 9.2%) and industry funding (11/185, 6%).

*Spin in abstracts*

Among the included systematic reviews, 31.9% (59/185) contained at least one form of spin in the abstract. Some articles contained more than one type of spin; therefore, 61 instances of spin were found. The most common form of spin found in the 185 articles was type 3 in 42 articles (42/185, 22.7%). Following type 3 spin, 10 articles (10/185, 5.4%) contained type 1 spin. Regarding type 4 spin, safety was not assessed in 145 systematic reviews. Therefore, this spin type was only evaluated in 40 of our included studies. Of those 40 studies, 15% (6/40) contained type 4 spin. Spin type 2 (2/185, 1.1%), type 5 (6/185, 3.2%), type 7 (1/185, 0.5%), and type 9 (3/185, 1.6%) were found less frequently. No abstracts contained type 6 or type 8 spin. There was no significance between spin and any of the evaluated study characteristics. All study characteristics in the presence of spin are mentioned in Table 2.

*AMSTAR-2*

After appraising the systematic reviews with the AMSTAR-2 instrument, we rated 110 (59.5%) of the 185 studies as "critically low," 29 (15.7%) as "low," 22 (11.9%) as "moderate," and 24 (13%) as "high." We found no statistical significance between spin and AMSTAR-2 appraisals (Table 3).

## Discussion

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Our study found spin in approximately one-third (31.9%) of the included systematic reviews. Spin has been shown to be prevalent in RCTs across many different fields of medicine [9,25]. Recent studies have shown spin to be present not only within RCT abstracts but also within abstracts of systematic reviews [26,27]. The results of the present study support the findings by Austin et al which identified one-third to one-half of systematic review abstracts contained at least one form of spin [25]. Although multiple studies have evaluated spin in systematic reviews, to our knowledge, ours is the first to evaluate the use of spin within antiplatelet research. Here, we highlight the most common and the most severe forms of spin we found and the influence they have on the clinical use of antiplatelet therapies.

The most common type of spin found among the systematic reviews and meta-analyses that we analyzed was type 3 spin (*selective reporting of or overemphasis on efficacy outcome or analysis favoring the beneficial effect of the experimental intervention)*. To further describe type 3 spin, if efficacy outcomes are specified within the full text and are not reported in the abstract, spin is committed. Consider the following example noted by Dundar et al [28]. They set out to examine mortality and major adverse effects associated with thrombolytic agents (alteplase, tenecteplase, streptokinase, and reteplase) in the treatment of acute myocardial infarction. In addition to mortality, their outcome measures included bleeding, stroke, reinfarction, allergy, and anaphylaxis. Of the 6 outcomes, only morality and stroke were reported in the abstract. Interestingly, 3 of the 4 outcomes that were not reported in the abstract (allergic reaction, anaphylaxis, and reinfarction) showed no statistical significance. Due to omission of these nonsignificant harm outcomes, the efficacy outcome of decreased mortality was selectively reported. Selectively reporting efficacy outcomes while leaving out harm outcomes can lead

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physicians to believe that an intervention is safer than it is. In this case of ACS, this belief could potentially lead to preventable adverse outcomes.

The most severe type of spin (*conclusion contains recommendations for clinical practice not supported by the findings*) was identified in our sample. This form of spin had the second greatest frequency among the included systematic reviews in our study. A study by Kinnaird et al provides an example of this form of spin. The authors claim that "This study suggests that bivalirudin is more effective and safer than heparin monotherapy and should, therefore, be preferred over heparin monotherapy" [29]. However, the recommendation for using bivalirudin over heparin was not supported by all outcomes of interest. Nonsignificant outcomes of interest included risk for stroke and myocardial infarction. Without consistent significant findings for all outcome measures, the claim for clinical practice is spin. In our view, this spin type is the most severe because a recommendation for clinical practice can have an immediate influence on patient care.

The methodological quality of the included systematic reviews was analyzed using AMSTAR-2. Among the reviews in our sample, over half were scored as critically low based on the AMSTAR-2 grading criteria. AMSTAR-2 defines this category as having at least one critical flaw rendering the systematic review unable to provide an accurate summary or recommendation [17]. However, the presence of spin in the abstract was not significantly associated with a low AMSTAR-2 score for the systematic review. In light of these findings, spin may be found in systematic reviews regardless of their quality; there may be a multitude of factors propagating spin aside from quality, including abstract word limitations, funding conflicts of interest, and

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lack of author awareness of spin. Although improving methodological quality may not eradicate the presence of spin, our findings support an opportunity for future systematic reviews to increase the quality of publications via adherence to AMSTAR-2.

Our study illustrates a current gap in transparency within the research pertaining to antiplatelet therapies. Regardless of whether it is intentional or not, the misrepresentation of data could lead to suboptimal patient care [30]. To help improve the reporting quality of systematic reviews on antiplatelet therapies following ACS, we have a few recommendations. First, readers must be made aware that spin exists within systematic reviews, especially as physicians often use systematic reviews to develop clinical practice guidelines. By creating an online resource dedicated to teaching spin, as well as presenting on spin at medical conferences, we believe knowledge of spin would be increased among medical students, residents, and physicians. Next, we suggest that journal editors require the implementation of spin screening prior to author submission. Our rationale is that entities who control article publications have the greatest influence to change the amount of spin present in publications. Lastly, PRISMA and PRISMA-A were designed to improve the reporting quality of systematic reviews. However, our study did not find less spin reported in publications that adhered to PRISMA. Thus, we suggest the addition of an item regarding spin to PRISMA and PRISMA-A to help reduce spin present in publications and systematic reviews.

**Strengths and Limitations**

Our study contained key strengths and limitations. Regarding strengths, we strictly adhered to our protocol, which was published prior to data extraction on OSF[12]. We made our protocol,

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data, analysis scripts, data extraction forms, and other study artifacts available online to increase our transparency and reproducibility. Furthermore, we used an independent group to verify our results and implemented both screening and data extraction in a masked, duplicate manner to strengthen our study and minimize bias. This process is recommended by the Cochrane Collaboration. Regarding limitations, spin by nature is subjective and interpretation could vary between extractors. First, the effect of spin on the reader's opinions and medical recommendations could not be judged. The degree of spin use may have affected readers differently. To help mitigate the subjectivity of spin, we used online coursework and training to prepare the team for screening and data extraction. An additional limitation is that our study had a cross-sectional design that did not allow for cause and effect analysis or generalizations to be made to other systematic reviews published in other journals. Furthermore, selection bias may have negatively influenced the data collection. However, this effect was reduced due to the extensive training on methodology and data collection performed prior to extraction. Lastly, our searches may not have returned all systematic reviews on antiplatelet therapies following ACS although we conducted our searches using the largest bibliographic databases (MEDLINE and Embase).

**Conclusion**

In conclusion, our study found spin is present in abstracts of systematic reviews regarding antiplatelet therapies following ACS. This misrepresentation of research data poses a threat to optimal evidence-based clinical use of antiplatelet therapy following ACS. There are steps that can be taken to improve the quality of antiplatelet therapy systematic reviews.

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**Figure Legend:**

**Table 1.** Spin types and frequencies in abstracts

**Table 2.** General characteristics of systematic reviews and meta-analyses

**Table 3.** AMSTAR-2 Items and Frequency of Responses

**Figure 1.** Search queries

**Figure 2.** PRISMA flow diagram of study selection

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**Table 1.** Spin types and frequencies in abstracts (n=185)

| Nine most severe types of spin | No. (%), containing spin |
|---|---|
| 1) Conclusion contains recommendations for clinical practice not supported by the findings. | 10 (5.4) |
| 2) Title claims or suggests a beneficial effect of the experimental intervention not supported by the findings. | 2 (1.1) |
| 3) Selective reporting of or overemphasis on efficacy outcomes or analysis favoring the beneficial effect of the experimental intervention. | 42 (22.7) |
| 4) Conclusion claims safety based on non-statistically significant results with a wide confidence interval. | 6 (15.0) |
| 5) Conclusion claims the beneficial effect of the experimental treatment despite high risk of bias in primary studies. | 6 (3.2) |
| 6) Selective reporting of or overemphasis on harm outcomes or analysis favoring the safety of the experimental intervention. | 0 (0) |
| 7) Conclusion extrapolates the review's findings to a different intervention (i.e., claiming efficacy of one specific intervention although the review covers a class of several interventions). | 1 (0.5) |
| 8) Conclusion extrapolates the review's findings from a surrogate marker or a specific outcome to the global improvement of the disease. | 0 (0) |
| 9) Conclusion claims the beneficial effect of the experimental treatment despite reporting bias. | 3 (1.6) |

*safety not assessed in 145, n= 40

only

**Table 2.** General characteristics of systematic reviews and meta-analyses

| Characteristics | No. (%) of Articles (n=185) | | Odds Ratio (95% CI) | |
|---|---|---|---|---|
| | Total (%) | Abstract Contains Spin | Unadjusted | Adjusted |
| Article mentions adherence to PRISMA | | | | |
| No | 106 (17.8) | 36 (19.5) | 1 [Reference] | 1 [Reference] |
| Yes | 79 (42.7) | 23 (12.4) | 0.67 (0.34-1.32) | 0.42 (0.16-1.07) |
| Publishing journal recommends adherence to PRISMA | | | | |
| No | 124 (67.0) | 43 (23.2) | 1 [Reference] | 1 [Reference] |
| Yes | 61 (33) | 16 (8.6) | 0.80 (0.43-1.50) | 0.65 (0.29-1.46) |
| Funding source | | | | |
| Not Funded | 44 (23.8) | 16 (8.6) | 1 [Reference] | 1 [Reference] |
| Industry | 11 (5.9) | 4 (2.2) | 1.00 (0.25-3.95) | 1.27 (0.23-7.04) |
| Not Mentioned | 93 (50.3) | 26 (14.1) | 0.68 (0.32-1.46) | 0.62 (0.23-1.64) |
| Private | 20 (10.8) | 7 (3.8) | 0.94 (0.31-2.85) | 0.86 (0.23-3.17) |
| Public | 17 (9.2) | 6 (3.2) | 0.95 (0.30-3.07) | 0.96 (0.24-3.76) |
| AMSTAR-2 Rating | | | | |
| High | 24 (13) | 7 (3.8) | 1 [Reference] | 1 [Reference] |
| Moderate | 110 (59.5) | 7 (3.8) | 1.13 (0.32-3.98) | 0.93 (0.23-3.87) |
| Low | 29 (15.7) | 12 (6.5) | 1.71 (0.54-5.41) | 1.68 (0.44-6.49) |
| Critically Low | 110 (59.5) | 33 (17.8) | 1.04 (0.39-2.75) | 1.19 (0.38-3.77) |
| Journal Impact Factor*, M (SD) | 7.80 (11.52) | 6.90 (8.45) | 0.99 (0.96-1.02) | 1.01 (0.97-1.05) |
| Publication Year (1991-2020) | | | 1.05 (1.00-1.11) | 1.09 (1.00-1.18) |

*18 journals did not have an impact factor

**Table 3.** AMSTAR-2 Items and Frequency of Responses (n=185)

| AMSTAR-2 Item | Response, n (%) | | |
|---|---|---|---|
| | Yes | No | Partial Yes |
| 1) Did the research questions and inclusion criteria for the review include the elements of PICO? | 185 (100) | 0 (0) | 0 (0) |
| 2) Did the report of the review contain an explicit statement that the review methods were established prior to the conduct of the review and did the report justify any significant deviations from the protocol? | 75 (40.5) | 108 (58.4) | 2 (1.1) |
| 3) Did the review authors explain their selection of the study designs for inclusion in the review? | 151 (81.6) | 25 (13.5) | 9 (4.9) |
| 4) Did the review authors use a comprehensive literature search strategy? | 55 (29.7) | 42 (22.7) | 88 (47.6) |
| 5) Did the review authors perform study selection in duplicate? | 128 (69.2) | 57 (30.8) | 0 (0) |
| 6) Did the review authors perform data extraction in duplicate? | 126 (68.1) | 59 (31.2) | 0 (0) |
| 7) Did the review authors provide a list of excluded studies and justify the exclusions? | 6 (3.2) | 51 (27.6) | 128 (69.1) |
| 8) Did the review authors describe the included studies in adequate detail? | 137 (74.1) | 12 (6.5) | 36 (19.4) |
| 9) Did the review authors use a satisfactory technique for assessing the risk of bias (RoB) in individual studies that were included in the review? | 53 (28.6) | 127 (68.6) | 5 (2.7) |
| 10) Did the review authors report on the sources of funding for the studies included in the review? | 7 (3.8) | 178 (96.2) | 0 (0) |
| 11) If meta-analysis was performed, did the review authors use appropriate methods for statistical combination of results?[a] | 142 (76.8) | 34 (18.4) | 0 (0) |
| 12) If meta-analysis was performed, did the review authors assess the potential impact of RoB in individual studies on the results of the meta-analysis or other evidence synthesis?[a] | 72 (38.9) | 103 (55.1) | 0 (0) |
| 13)Did the review authors account for RoB in primary studies when interpreting/discussing the results of the review? | 74 (40.0) | 111 (60.0) | 0 (0) |
| 14) Did the review authors provide a satisfactory explanation for, and discussion of, any heterogeneity observed in the results of the review? | 145 (78.4) | 40 (21.6) | 0 (0) |
| 15) If they performed quantitative synthesis did the review authors carry out an adequate investigation of publication bias (small study bias) and discuss its likely impact on the results of the review?[a] | 110 (59.5) | 66 (35.7) | 0 (0) |
| 16) Did the review authors report any potential sources of conflict of interest, including any funding they received for conducting the review | 127 (68.6) | 58 (31.4) | 0 (0) |

[a]9 articles did not perform a meta-analysis, thus n=176

| Ovid MEDLINE: | Ovid Embase: |
|---|---|
| 1. exp Platelet Aggregation Inhibitors/<br>2. (platelet* adj2 (antagonist* or inhibit* or antiaggregant*)).mp.<br>3. ((antiplatelet* or antithrombocyt*) adj2 (treatment* or therap* or agent* or drug*)).mp.<br>4. 1 or 2 or 3<br>5. exp "Systematic Review"/<br>6. exp Meta-Analysis/<br>7. ("systematic review" or "meta-analysis" or (systematic* adj1 review*)).ti,ab.<br>8. 5 or 6 or 7<br>9. 4 and 8 | 1. exp antithrombocytic agent/<br>2. (platelet* adj2 (antagonist* or inhibit* or antiaggregant*)).mp.<br>3. ((antiplatelet* or antithrombocyt*) adj2 (treatment* or therap* or agent* or drug*)).mp.<br>4. 1 or 2 or 3<br>5. exp "systematic review"/<br>6. exp meta analysis/<br>7. ("systematic review" or "meta-analysis" or (systematic* adj1 review*)).ti,ab.<br>8. 5 or 6 or 7<br>9. 4 and 8 |

**Figure 1.** Search queries

215x279mm (150 x 150 DPI)

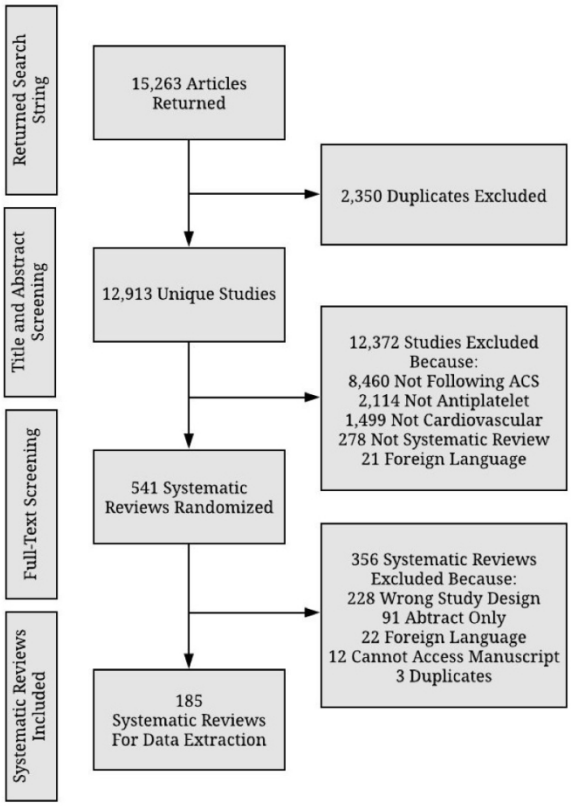



**Figure 2.** PRISMA flow diagram of study selection

215x279mm (150 x 150 DPI)

STROBE Statement—Checklist of items that should be included in reports of ***cross-sectional studies***

| | Item No | Recommendation | Page No |
|---|---|---|---|
| **Title and abstract** | 1 | (*a*) Indicate the study's design with a commonly used term in the title or the abstract | 0 |
| | | (*b*) Provide in the abstract an informative and balanced summary of what was done and what was found | 1 |
| **Introduction** | | | |
| Background/rationale | 2 | Explain the scientific background and rationale for the investigation being reported | 2 |
| Objectives | 3 | State specific objectives, including any prespecified hypotheses | 3 |
| **Methods** | | | |
| Study design | 4 | Present key elements of study design early in the paper | 3 |
| Setting | 5 | Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection | 4 |
| Participants | 6 | (*a*) Give the eligibility criteria, and the sources and methods of selection of participants | 4 |
| Variables | 7 | Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable | 5 |
| Data sources/ measurement | 8* | For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group | 6 |
| Bias | 9 | Describe any efforts to address potential sources of bias | 4 |
| Study size | 10 | Explain how the study size was arrived at | 4 |
| Quantitative variables | 11 | Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why | 6 |
| Statistical methods | 12 | (*a*) Describe all statistical methods, including those used to control for confounding | 6 |
| | | (*b*) Describe any methods used to examine subgroups and interactions | 6 |
| | | (*c*) Explain how missing data were addressed | 7 |
| | | (*d*) If applicable, describe analytical methods taking account of sampling strategy | 6 |
| | | (*e*) Describe any sensitivity analyses | 6 |
| **Results** | | | |
| Participants | 13* | (a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed | 7 |
| | | (b) Give reasons for non-participation at each stage | 7 |
| | | (c) Consider use of a flow diagram | 25 |
| Descriptive data | 14* | (a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders | 7 |
| | | (b) Indicate number of participants with missing data for each variable of interest | 7 |
| Outcome data | 15* | Report numbers of outcome events or summary measures | 8 |
| Main results | 16 | (*a*) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included | 8 |

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| | | | |
|---|---|---|---|
| | | (*b*) Report category boundaries when continuous variables were categorized | 8 |
| | | (*c*) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period | 8 |
| Other analyses | 17 | Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses | 8 |
| **Discussion** | | | |
| Key results | 18 | Summarise key results with reference to study objectives | 9 |
| Limitations | 19 | Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias | 12 |
| Interpretation | 20 | Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence | 10 |
| Generalisability | 21 | Discuss the generalisability (external validity) of the study results | 12 |
| **Other information** | | | |
| Funding | 22 | Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based | 1 |

*Give information separately for exposed and unexposed groups.

**Note:** An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at http://www.plosmedicine.org/, Annals of Internal Medicine at http://www.annals.org/, and Epidemiology at http://www.epidem.com/). Information on the STROBE Initiative is available at www.strobe-statement.org.

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BMJ Open

# Spin within systematic review abstracts on antiplatelet therapies after acute coronary syndrome: A cross-sectional study

| | |
|---|---|
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**Title:** Spin within systematic review abstracts on antiplatelet therapies after acute coronary syndrome: A cross-sectional study

**Short Title:** Spin in systematic reviews regarding antiplatelet therapies

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**Contributorship Statement:**
Audrey Wise: acquisition, analysis and interpretation of data, manuscript drafting and revision
Deepika Mannem: acquisition, analysis and interpretation of data, manuscript drafting
Wade Arthur: conception and design of the work, drafting and revision of manuscript
Ryan Ottwell: conception and design of the work, revision of manuscript
Benjamin Greiner: drafting of the manuscript
Derek Srouji: revision of manuscript
Daniel Wildes: revision of manuscript
Micah Hartwell: analysis and interpretation of data, revision of manuscript
Drew N Wright: conceptual design of the work
Jam Khojasteh: analysis and interpretation of data
Matthew Vassar: conceptual design of the work, revision of manuscript

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## Abstract

**Objectives:** Spin is a reporting practice in which study results are misrepresented by overestimating efficacy or underestimating harm. Prevalence of spin varies between clinical specialties, and estimates are based almost entirely on clinical trials. Little is known about spin in systematic reviews.

**Design:** We performed a cross-sectional analysis searching MEDLINE and Embase for systematic reviews and meta-analyses pertaining to antiplatelet therapies following acute coronary syndrome on June 2, 2020. Data were extracted evaluating the presence of spin and study characteristics, including methodological quality as rated by AMSTAR-2. All data extraction was conducted in a masked, duplicate manner from June 2, 2020 to June 26, 2020.

**Participants and Setting:** Not applicable.

**Primary and Secondary Outcome Measures:** We assessed abstracts of systematic reviews on antiplatelet therapy following acute coronary syndrome and evaluated the prevalence of the 9 most severe types of spin. We additionally explored associations between spin and certain study characteristics, including quality.

**Results:** Our searches returned 15,263 articles, and 185 systematic reviews met inclusion criteria. Of these 185 reviews, 31.9% (59/185) contained some form of spin in the abstract. Seven forms of spin (1, 2, 3, 4, 5, 7, and 9) among the 9 most severe were identified. No instances of types 6 or 8 were found. There were no statistically significant relationships between spin and the evaluated study characteristics or AMSTAR-2 appraisals.

**Conclusions:** Spin was present in abstracts for systematic reviews and meta-analyses; subsequent studies are needed to identify correlations between spin and specific study


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characteristics. There were no statistically significant associations between spin and study characteristics or AMSTAR-2 ratings; however, implementing changes will ensure that spin is reduced in the field of cardiology as well as other fields of medicine.

## Article Summary

### Strengths and Limitations of this Study:

- We strictly adhered to our protocol, which was published prior to data extraction on Open Science Framework. We made our protocol, data, analysis scripts, data extraction forms, and other study artifacts available online to increase our transparency and reproducibility.
- As recommended by Cochrane Collaboration, we used an independent group to verify our results and implemented both screening and data extraction in a masked, duplicate manner to strengthen our study and minimize bias.
- The effect of spin on the reader's opinions and medical recommendations could not be judged, and the degree of spin use may have affected readers differently. To help mitigate the subjectivity of spin, we used online coursework and training to prepare our team for screening and data extraction.
- Our study had a cross-sectional design that did not allow for cause and effect analysis or generalizations to be made to other systematic reviews published in other journals.
- Selection bias may have negatively influenced the data collection. However, this effect was reduced due to the extensive training on methodology and data collection performed prior to extraction.

### Funding Statement:

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**Competing Interest:**

None declared.

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## Introduction

As the gold standard in evidence-based medicine, regulating bodies often use systematic reviews to develop clinical practice guidelines that physicians refer to for treatment decisions [1,2]. Hence, the findings contained in systematic reviews and in their abstracts directly influence the care that patients receive and their prognosis.

The systematic reviews themselves serve as a means to collate all relevant primary data on a specific topic, critically appraise the literature, and condense the findings into a useful synthesis. A major issue for patient care is that the reviews often contain outcome reporting bias; that is, they share only positive findings, show selection bias, and/or transfer bias from primary data [3]. This bias is further complicated by clinicians having limited access to full-text articles, as well as limited time to read them in decision-making contexts. A 2013 study, for example, found that nearly 75% of physicians read the abstract alone, rather than the full-text of the article [4]. Importantly, abstracts have been found to not always accurately represent the findings reported in articles. For example, Toma and colleagues reported that 41% of the abstracts of randomized controlled trials contained different efficacy estimates than the full text [5]. Consistent with that report, other studies have shown that over 50% of abstracts had discrepancies in the results compared with the full text, and only 50% of abstracts discussed the side effects or harms experienced during clinical trials [6,7]. Considering the importance of evidence-based medicine, the necessity for reducing bias, especially in the abstracts of systematic reviews, cannot be overstated.

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One form of bias that is of increasing interest to systematic reviewers, clinical trial researchers, and epidemiologists is spin. Spin involves misrepresenting research results to mislead readers into favoring a certain outcome. It has been identified in multiple studies, particularly in clinical trial research, accessible from these publicly available datasets [8,9].

Accurate and objective research findings are necessary to guide patient care. For example, patients who have experienced acute coronary syndrome (ACS) with subsequent initiation of antiplatelet therapy are particularly at risk for morbidity and mortality arising from antiplatelet therapy inaccuracies. If clinicians misinterpret (or are misled by) clinical research on antiplatelet therapy, the results could be deadly.

However, no study has analyzed spin in this body of literature. Thus, the primary objective of this study was to analyze the presence of spin in the abstracts of systematic reviews on antiplatelet therapy following ACS. The secondary objective was to assess the study characteristics, including journal impact factor, funding source, and article type, associated with spin. We additionally describe methods for reducing spin in antiplatelet publications.

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## Methods

*Search Strategy*

A systematic review librarian (DW) constructed the search strategies for the MEDLINE (Ovid) and Embase (Ovid) databases to locate systematic reviews and meta-analyses pertaining to antiplatelet therapies (Figure 1). DW conducted these searches on June 2, 2020. Following the execution of these search strategies, the results were combined and uploaded into a screening platform, Rayyan (https://rayyan.qcri.org/). The duplicates were removed, and the titles and abstracts were independently screened for eligibility by 2 researchers (AW and DM) [10]. This process was performed in a masked, duplicate manner. Once the initial screening was complete, AW and DM resolved disagreements through discussion.

*Eligibility Criteria*

The following inclusion criteria had to be met for an article to be eligible for this study. The article must (1) be a systematic review with or without a meta-analysis; (2) focus on antiplatelet, anticoagulation, or thrombolytic therapies following ACS; (3) be conducted on human subjects only; (4) be available in English; and (5) have an abstract. The following study types were excluded from our study: observational studies (case-control, cohort, surveys), clinical trials, narrative reviews, and systematic reviews not related to antiplatelet therapies. We also excluded duplicates, withdrawn or retracted studies, letters to the editor, and any remaining article that did not meet the inclusion criteria. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA) definitions of systematic reviews and meta-analyses were used [11]. The selected studies were then uploaded to STATA for randomization, and as long as the previous criteria were met, the first 200 articles were sequentially extracted.

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*Training*

Several levels of training were completed to ensure the quality of this study. Prior to the start of the study, all researchers were trained on an overview of study designs in clinical research, deduplication, and screening on Rayyan, and publication of protocols using Open Science Framework (OSF) [12]. The 2 researchers who conducted the screening (AW and DM) also completed an online training course on systematic reviews and meta-analyses offered by John Hopkins University on the Coursera platform [13]. Following the course, the researchers undertook 2 days of online and in-person training. The training session covered the definition and interpretation of the 9 most severe forms of spin in systematic review abstracts as defined by Yavchitz et al [14]. Finally, data extractors were trained on how to analyze a systematic review via A MeaSurement Tool to Assess systematic Reviews (AMSTAR-2). The AMSTAR tool was used to determine the methodological quality of each included systematic review and meta-analysis. A detailed outline of the training can be found in our study protocol.

*Identification of Spin and Data Extraction*

Data extraction was completed by AW and DM, using a pilot-tested Google form to evaluate for 9 types of spin. These forms of spin as defined in Yavchitz et al[14] studies were chosen and settled on following a Q-sort survey with members of the Cochrane Collaboration. This consisted of method review and bias method group members from over 400 invited participants. Members ranked examples on the Q-sort survey for more severe and prevalent forms of bias and were evaluated in our study. The selection of the 9 types of spin in our study is derived from the most severe types of spin Yavchitz el al([14]) identified as well as from publicly available data in

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recent published literature[15] on spin for completeness. The following general information was extracted from all reviews: the intervention type (surgery, pharmacologic, and nonpharmacologic); the year in which the review was received by the journal; the funding source (industry, private, public, none, not-mentioned, and hospital); whether the review discussed adherence to PRISMA [16] or PRISMA for abstracts [17]; whether the journal recommended adherence to PRISMA; and the journal's 5-year impact factor. The 9 types of spin are defined in Table 1. Researchers coded each of the 9 spin types as being either present or absent from the systematic review, except for type 4 spin if safety was not addressed in the manuscript. If a systematic review did not address safety, it was not included in the analysis for type 4 spin. Any disagreements about the presence of spin were reconciled between AW and DM, and if an agreement could not be reached WA and MV acted as mediators. Following spin data extraction, the quality of each systematic review was analyzed using AMSTAR-2 (https://amstar.ca/) [18]. AMSTAR-2 has previously been used to evaluate systematic reviews in multiple medical disciplines, including surgery, cancer, psychiatry, pain management, and general medical interventions [18–22]. Based on scores from the 16-item scale, reviews were given confidence ratings of high, moderate, low, or critically low [18]. Interrater reliability scores (assessed by kappa) for AMSTAR-2 ranged from .31 to 1.0 [18], with additional studies supporting its reliability, finding moderate interrater reliability (median K = 0.56) among multiple raters. Construct validity coefficients were high (AMSTAR, r = 0.91; Risk of Bias in Systematic Reviews [ROBIS], r = 0.84) [23]. All study characteristics, spin data, and AMSTAR-2 data were extracted in a masked, duplicate manner [14].

*Statistical Analysis*

The overall frequency of spin, study characteristics, and AMSTAR-2 ratings were analyzed using descriptive statistics, and the results are reported as both frequency counts and percentages. Following the screening, we included 185 systematic reviews. Binary logistic regression was used to determine the strength of associations via odds ratios between the occurrence of spin in a study's abstract with journal impact factor, study characteristics, and AMSTAR-2 appraisal rating, and a multivariable model was used to control for multiple characteristics. Prior to the start of the study, we determined that 185 systematic reviews were needed to power the analyses using GPower 3.1.9.7. Our analytic decisions are included in our protocol, and Stata 16.1 (StataCorp, LLC, College Station, TX) was used for all analyses.

*Patient and Public Involvement*

This study did not include any human participants or public involvement and therefore did not meet the regulatory definition of human subject research per the U.S. Code of Federal Regulations and was not subject to institutional review board oversight.

*Oversight, Data Availability, Transparency, Reproducibility, and Reporting*

To ensure that our study is transparent and reproducible, we have made the protocol, extraction forms, data analysis scripts, and other study artifacts available via OSF. Our data and analysis scripts were sent to an independent investigation team and reanalyzed in a masked fashion to further the reproducibility. This study was conducted in tandem with other studies evaluating the presence of spin in systematic reviews in other medical fields. These studies adhered to a common methodology, which has been described elsewhere. While drafting this article, we

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integrated the relevant reporting guidelines from PRISMA [24] and Murad and Wang's [25] guidelines for meta-epidemiological studies.

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## Results

*General Characteristics*

Our search queries returned 15,263 articles and 12,913 remained after deduplication. Of these, 12,372 were excluded after the initial title and abstract screening, which left 541 unique systematic reviews. A full-text review of the articles further excluded 356 reviews. Finally, 185 systematic reviews and meta-analyses pertaining to antiplatelet therapies following ACS remained and were included in data extraction. Our exclusion list from the screening process with rationale can be found in Figure 2.

All studies meeting inclusion criteria analyzed pharmacologic interventions. Of the 185 reviews, 61 (32.9%) systematic reviews were published in journals that recommended adherence to PRISMA, and 79 (42.7%) of the papers mentioned adherence to PRISMA or PRISMA-A. The majority of the systematic reviews and meta-analyses did not mention a funding source (93/185, 50.2%), followed by systematic reviews that did not receive any kind of funding (44/185, 23.8%). Of the papers that mentioned a funding source (92/185, 49.7%), the most common funding source was private funding (20/185, 10.8%), followed by public funding (17/185, 9.2%) and industry funding (11/185, 6%).

*Spin in abstracts*

Among the included systematic reviews, 31.9% (59/185) contained at least one form of spin in the abstract. Some articles contained more than one type of spin; therefore, 61 instances of spin were found. The most common form of spin found in the 185 articles was type 3 in 42 articles (42/185, 22.7%). Following type 3 spin, 10 articles (10/185, 5.4%) contained type 1 spin.

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Regarding type 4 spin, safety was not assessed in 145 systematic reviews. Therefore, this spin type was only evaluated in 40 of our included studies. Of those 40 studies, 15% (6/40) contained type 4 spin. Spin type 2 (2/185, 1.1%), type 5 (6/185, 3.2%), type 7 (1/185, 0.5%), and type 9 (3/185, 1.6%) were found less frequently. No abstracts contained type 6 or type 8 spin. There was no significance between spin and any of the evaluated study characteristics. All study characteristics in the presence of spin are mentioned in Table 2.

*AMSTAR-2*

After appraising the systematic reviews with the AMSTAR-2 instrument, we rated 110 (59.5%) of the 185 studies as "critically low," 29 (15.7%) as "low," 22 (11.9%) as "moderate," and 24 (13%) as "high." We found no statistical significance between spin and AMSTAR-2 appraisals (Table 3).

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## Discussion

Our study found spin in approximately one-third (31.9%) of the included systematic reviews. Spin has been shown to be prevalent in RCTs across many different fields of medicine [9,26]. Recent studies have shown spin to be present not only within RCT abstracts but also within abstracts of systematic reviews [27,28]. The results of the present study support the findings by Austin et al which identified one-third to one-half of systematic review abstracts contained at least one form of spin, noted in their publicly available dataset [26]. Although multiple studies have evaluated spin in systematic reviews, to our knowledge, ours is the first to evaluate the use of spin within antiplatelet research. Here, we highlight the most common and the most severe forms of spin we found and the influence they have on the clinical use of antiplatelet therapies.

The most common type of spin found among the systematic reviews and meta-analyses that we analyzed was type 3 spin (*selective reporting of or overemphasis on efficacy outcome or analysis favoring the beneficial effect of the experimental intervention)*. To further describe type 3 spin, if efficacy outcomes are specified within the full text and are not reported in the abstract, spin is committed. Consider the following example noted by Dundar et al [29]. They examined mortality and major adverse effects associated with thrombolytic agents (alteplase, tenecteplase, streptokinase, and reteplase) in the treatment of acute myocardial infarction. In addition to mortality, their outcome measures included bleeding, stroke, reinfarction, allergy, and anaphylaxis. Of the 6 outcomes, only morality and stroke were reported in the abstract. Interestingly, 3 of the 4 outcomes that were not reported in the abstract (allergic reaction, anaphylaxis, and reinfarction) showed no statistical significance. Due to omission of these nonsignificant harm outcomes, the efficacy outcome of decreased mortality was selectively

reported. Selectively reporting efficacy outcomes while leaving out harm outcomes can lead physicians to believe that an intervention is safer than it is; this belief may potentially lead to preventable adverse outcomes.

The most severe type of spin (*conclusion contains recommendations for clinical practice not supported by the findings*) was identified in our sample. This form of spin had the second greatest frequency among the included systematic reviews in our study. A study by Kinnaird et al provides an example of this form of spin. The authors claim that "This study suggests that bivalirudin is more effective and safer than heparin monotherapy and should, therefore, be preferred over heparin monotherapy" [30]. However, the recommendation for using bivalirudin over heparin was not supported by all outcomes of interest. Nonsignificant outcomes of interest included risk for stroke and myocardial infarction. Without consistent significant findings for all outcome measures, the claim for clinical practice is spin. In our view, this spin type is the most severe because a recommendation for clinical practice can have an immediate influence on patient care.

The methodological quality of the included systematic reviews was analyzed using AMSTAR-2. Among the reviews in our sample, over half were scored as critically low based on the AMSTAR-2 grading criteria. AMSTAR-2 defines this category as having at least one critical flaw rendering the systematic review unable to provide an accurate summary or recommendation [18]. However, the presence of spin in the abstract was not significantly associated with a low AMSTAR-2 score for the systematic review. In light of these findings, spin may be found in systematic reviews regardless of their quality; there may be a multitude of factors propagating

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spin aside from quality, including abstract word limitations, funding conflicts of interest, and lack of author awareness of spin. Although improving methodological quality may not eradicate the presence of spin, our findings support an opportunity for future systematic reviews to increase the quality of publications via adherence to AMSTAR-2.

Our study illustrates a current gap in transparency within the research pertaining to antiplatelet therapies. Whether intentional or not, motivating factors for the use of spin within manuscripts include confirmation bias, allegiance bias, and the misrepresentation of results in hopes for publication. Ioannidis et al suggest that the motives include a belief that positive results are associated with an increased likelihood of publication and greater clinical significance [31]. Regardless of whether it is intentional or not, the misrepresentation of data may lead to suboptimal patient care [32]. To help improve the reporting quality of systematic reviews on antiplatelet therapies following ACS, we have a few recommendations. First, readers must be made aware that spin exists within systematic reviews, especially as physicians often use systematic reviews to develop clinical practice guidelines. Given that Bero et al identified in their publicly available dataset that spin is less prevalent in publications evaluating spin [33], we believe an awareness of spin is vital to decreasing spin in publications. Thus, by creating an online resource dedicated to teaching spin, as well as presenting on spin at medical conferences, we believe knowledge of spin would be increased among medical students, residents, and physicians and its prevalence in publications may decrease. Next, we suggest that journal editors require the implementation of spin screening prior to author submission. Our rationale is that entities who control article publications have the greatest influence to change the amount of spin present in publications. Furthermore, in an effort to deescalate spin, we urge journal editors to

strive for inferential reproducibility by using a second discussant to verify study findings prior to publication [31]. Inferential reproducibility is attained for a publication by an independent reviewer re-analyzing the study to see if the same results and conclusions are reached [34]. Additionally, we recommend journal editors emphasize the publication of studies with neutral results given that researchers believe positive results are more likely to be published [34]. Lastly, PRISMA and PRISMA-A were designed to improve the reporting quality of systematic reviews. However, our study did not find less spin reported in publications that adhered to PRISMA. Thus, we suggest the addition of an item regarding spin to PRISMA and PRISMA-A to help reduce spin present in publications and systematic reviews.

**Strengths and Limitations**

Our study contained key strengths and limitations. Regarding strengths, we strictly adhered to our protocol, which was published prior to data extraction on OSF. We made our protocol, data, analysis scripts, data extraction forms, and other study artifacts available online to increase our transparency and reproducibility. Furthermore, we used an independent group to verify our results and implemented both screening and data extraction in a masked, duplicate manner to strengthen our study and minimize bias. This process is recommended by the Cochrane Collaboration. Regarding limitations, spin by nature is subjective and interpretation could vary between extractors. First, the effect of spin on the reader's opinions and medical recommendations could not be judged. The degree of spin use may have affected readers differently. To help mitigate the subjectivity of spin, we used online coursework and training to prepare the team for screening and data extraction. An additional limitation is that our study had a cross-sectional design that did not allow for cause and effect analysis or generalizations to be

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made to other systematic reviews published in other journals. Furthermore, selection bias may have negatively influenced the data collection. However, this effect was reduced due to the extensive training on methodology and data collection performed prior to extraction. Lastly, our searches may not have returned all systematic reviews on antiplatelet therapies following ACS although we conducted our searches using the largest bibliographic databases (MEDLINE and Embase).

## Conclusion

In conclusion, our study found spin is present in abstracts of systematic reviews regarding antiplatelet therapies following ACS. Although there were no statistically significant relationships between spin and the evaluated study characteristics or AMSTAR-2 appraisals, this misrepresentation of research data poses a threat to optimal evidence-based clinical use of antiplatelet therapy following ACS. There are steps that can be taken to improve the quality of antiplatelet therapy systematic reviews.

## Ethics Statement

Our study did not contain human subjects, and therefore patient consent form was not applicable. Our study did not contain animal subjects, and therefore patient consent form was not applicable.

## Data Availability Statement

Regarding data availability, data are available in a public, open access repository and any data not on OSF are available upon reasonable request. To ensure that our study is transparent and reproducible, we have made the protocol, extraction forms, data analysis scripts, and other study artifacts available via OSF.

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**Figure Legend:**

**Table 1.** Spin types and frequencies in abstracts

**Table 2.** General characteristics of systematic reviews and meta-analyses

**Table 3.** AMSTAR-2 Items and Frequency of Responses

**Figure 1.** Search queries

**Figure 2.** PRISMA flow diagram of study selection

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| **Table 1.** Spin types and frequencies in abstracts (n=185) | |
|---|---|
| **Nine most severe types of spin** | **No. (%), containing spin** |
| 1) Conclusion contains recommendations for clinical practice not supported by the findings. | 10 (5.4) |
| 2) Title claims or suggests a beneficial effect of the experimental intervention not supported by the findings. | 2 (1.1) |
| 3) Selective reporting of or overemphasis on efficacy outcomes or analysis favoring the beneficial effect of the experimental intervention. | 42 (22.7) |
| 4) Conclusion claims safety based on non-statistically significant results with a wide confidence interval. | 6 (15.0) |
| 5) Conclusion claims the beneficial effect of the experimental treatment despite high risk of bias in primary studies. | 6 (3.2) |
| 6) Selective reporting of or overemphasis on harm outcomes or analysis favoring the safety of the experimental intervention. | 0 (0) |
| 7) Conclusion extrapolates the review's findings to a different intervention (i.e., claiming efficacy of one specific intervention although the review covers a class of several interventions). | 1 (0.5) |
| 8) Conclusion extrapolates the review's findings from a surrogate marker or a specific outcome to the global improvement of the disease. | 0 (0) |
| 9) Conclusion claims the beneficial effect of the experimental treatment despite reporting bias. | 3 (1.6) |
| *safety not assessed in 145, n= 40 | |

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**Table 2.** General characteristics of systematic reviews and meta-analyses

| Characteristics | No. (%) of Articles (n=185) | | Odds Ratio (95% CI) | |
|---|---|---|---|---|
| | Total (%) | Abstract Contains Spin | Unadjusted | Adjusted |
| Article mentions adherence to PRISMA | | | | |
| No | 106 (17.8) | 36 (19.5) | 1 [Reference] | 1 [Reference] |
| Yes | 79 (42.7) | 23 (12.4) | 0.67 (0.34-1.32) | 0.42 (0.16-1.07) |
| Publishing journal recommends adherence to PRISMA | | | | |
| No | 124 (67.0) | 43 (23.2) | 1 [Reference] | 1 [Reference] |
| Yes | 61 (33) | 16 (8.6) | 0.80 (0.43-1.50) | 0.65 (0.29-1.46) |
| Funding source | | | | |
| Not Funded | 44 (23.8) | 16 (8.6) | 1 [Reference] | 1 [Reference] |
| Industry | 11 (5.9) | 4 (2.2) | 1.00 (0.25-3.95) | 1.27 (0.23-7.04) |
| Not Mentioned | 93 (50.3) | 26 (14.1) | 0.68 (0.32-1.46) | 0.62 (0.23-1.64) |
| Private | 20 (10.8) | 7 (3.8) | 0.94 (0.31-2.85) | 0.86 (0.23-3.17) |
| Public | 17 (9.2) | 6 (3.2) | 0.95 (0.30-3.07) | 0.96 (0.24-3.76) |
| AMSTAR-2 Rating | | | | |
| High | 24 (13) | 7 (3.8) | 1 [Reference] | 1 [Reference] |
| Moderate | 110 (59.5) | 7 (3.8) | 1.13 (0.32-3.98) | 0.93 (0.23-3.87) |
| Low | 29 (15.7) | 12 (6.5) | 1.71 (0.54-5.41) | 1.68 (0.44-6.49) |
| Critically Low | 110 (59.5) | 33 (17.8) | 1.04 (0.39-2.75) | 1.19 (0.38-3.77) |
| Journal Impact Factor*, M (SD) | 7.80 (11.52) | 6.90 (8.45) | 0.99 (0.96-1.02) | 1.01 (0.97-1.05) |
| Publication Year (1991-2020) | | | 1.05 (1.00-1.11) | 1.09 (1.00-1.18) |

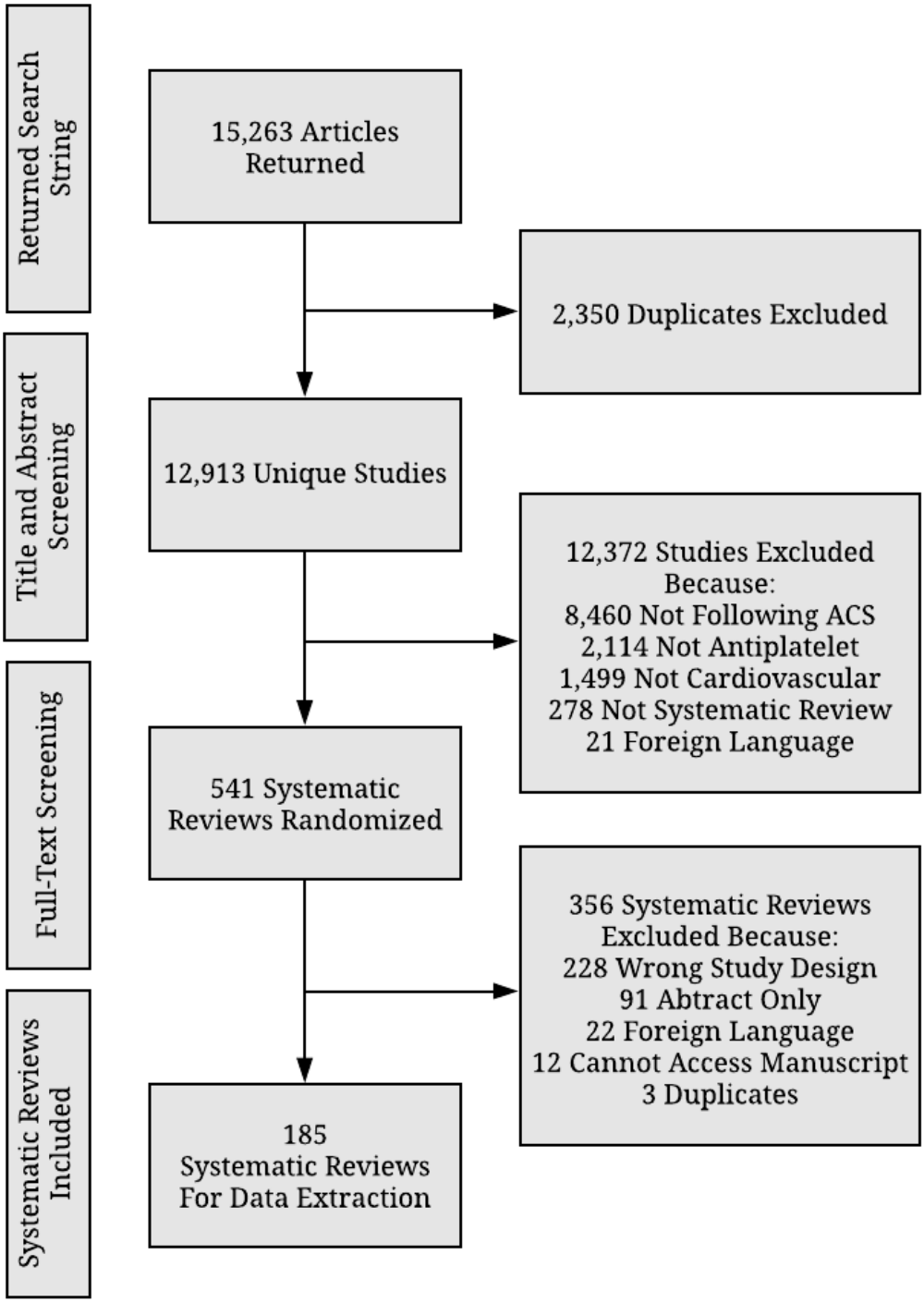
*18 journals did not have an impact factor

**Table 3.** AMSTAR-2 Items and Frequency of Responses (n=185)

| AMSTAR-2 Item | Response, n (%) | | |
|---|---|---|---|
| | Yes | No | Partial Yes |
| 1) Did the research questions and inclusion criteria for the review include the elements of PICO? | 185 (100) | 0 (0) | 0 (0) |
| 2) Did the report of the review contain an explicit statement that the review methods were established prior to the conduct of the review and did the report justify any significant deviations from the protocol? | 75 (40.5) | 108 (58.4) | 2 (1.1) |
| 3) Did the review authors explain their selection of the study designs for inclusion in the review? | 151 (81.6) | 25 (13.5) | 9 (4.9) |
| 4) Did the review authors use a comprehensive literature search strategy? | 55 (29.7) | 42 (22.7) | 88 (47.6) |
| 5) Did the review authors perform study selection in duplicate? | 128 (69.2) | 57 (30.8) | 0 (0) |
| 6) Did the review authors perform data extraction in duplicate? | 126 (68.1) | 59 (31.2) | 0 (0) |
| 7) Did the review authors provide a list of excluded studies and justify the exclusions? | 6 (3.2) | 51 (27.6) | 128 (69.1) |
| 8) Did the review authors describe the included studies in adequate detail? | 137 (74.1) | 12 (6.5) | 36 (19.4) |
| 9) Did the review authors use a satisfactory technique for assessing the risk of bias (RoB) in individual studies that were included in the review? | 53 (28.6) | 127 (68.6) | 5 (2.7) |
| 10) Did the review authors report on the sources of funding for the studies included in the review? | 7 (3.8) | 178 (96.2) | 0 (0) |
| 11) If meta-analysis was performed, did the review authors use appropriate methods for statistical combination of results?[a] | 142 (76.8) | 34 (18.4) | 0 (0) |
| 12) If meta-analysis was performed, did the review authors assess the potential impact of RoB in individual studies on the results of the meta-analysis or other evidence synthesis?[a] | 72 (38.9) | 103 (55.1) | 0 (0) |
| 13)Did the review authors account for RoB in primary studies when interpreting/discussing the results of the review? | 74 (40.0) | 111 (60.0) | 0 (0) |
| 14) Did the review authors provide a satisfactory explanation for, and discussion of, any heterogeneity observed in the results of the review? | 145 (78.4) | 40 (21.6) | 0 (0) |
| 15) If they performed quantitative synthesis did the review authors carry out an adequate investigation of publication bias (small study bias) and discuss its likely impact on the results of the review?[a] | 110 (59.5) | 66 (35.7) | 0 (0) |
| 16) Did the review authors report any potential sources of conflict of interest, including any funding they received for conducting the review | 127 (68.6) | 58 (31.4) | 0 (0) |

[a]9 articles did not perform a meta-analysis, thus n=176

| **Ovid MEDLINE:** | **Ovid Embase:** |
|---|---|
| 1. exp Platelet Aggregation Inhibitors/<br>2. (platelet* adj2 (antagonist* or inhibit* or antiaggregant*)).mp.<br>3. ((antiplatelet* or antithrombocyt*) adj2 (treatment* or therap* or agent* or drug*)).mp.<br>4. 1 or 2 or 3<br>5. exp "Systematic Review"/<br>6. exp Meta-Analysis/<br>7. ("systematic review" or "meta-analysis" or (systematic* adj1 review*)).ti,ab.<br>8. 5 or 6 or 7<br>9. 4 and 8 | 1. exp antithrombocytic agent/<br>2. (platelet* adj2 (antagonist* or inhibit* or antiaggregant*)).mp.<br>3. ((antiplatelet* or antithrombocyt*) adj2 (treatment* or therap* or agent* or drug*)).mp.<br>4. 1 or 2 or 3<br>5. exp "systematic review"/<br>6. exp meta analysis/<br>7. ("systematic review" or "meta-analysis" or (systematic* adj1 review*)).ti,ab.<br>8. 5 or 6 or 7<br>9. 4 and 8 |


Returned Search String
15,263 Articles Returned
2,350 Duplicates Excluded
Title and Abstract Screening
12,913 Unique Studies
12,372 Studies Excluded Because:
8,460 Not Following ACS
2,114 Not Antiplatelet
1,499 Not Cardiovascular
278 Not Systematic Review
21 Foreign Language
Full-Text Screening
541 Systematic Reviews Randomized
356 Systematic Reviews Excluded Because:
228 Wrong Study Design
91 Abtract Only
22 Foreign Language
12 Cannot Access Manuscript
3 Duplicates
Systematic Reviews Included
185 Systematic Reviews For Data Extraction

STROBE Statement—Checklist of items that should be included in reports of ***cross-sectional studies***

| | Item No | Recommendation | Page No |
|---|---|---|---|
| **Title and abstract** | 1 | (*a*) Indicate the study's design with a commonly used term in the title or the abstract | 0 |
| | | (*b*) Provide in the abstract an informative and balanced summary of what was done and what was found | 1 |
| **Introduction** | | | |
| Background/rationale | 2 | Explain the scientific background and rationale for the investigation being reported | 2 |
| Objectives | 3 | State specific objectives, including any prespecified hypotheses | 3 |
| **Methods** | | | |
| Study design | 4 | Present key elements of study design early in the paper | 3 |
| Setting | 5 | Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection | 4 |
| Participants | 6 | (*a*) Give the eligibility criteria, and the sources and methods of selection of participants | 4 |
| Variables | 7 | Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable | 5 |
| Data sources/ measurement | 8* | For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group | 6 |
| Bias | 9 | Describe any efforts to address potential sources of bias | 4 |
| Study size | 10 | Explain how the study size was arrived at | 4 |
| Quantitative variables | 11 | Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why | 6 |
| Statistical methods | 12 | (*a*) Describe all statistical methods, including those used to control for confounding | 6 |
| | | (*b*) Describe any methods used to examine subgroups and interactions | 6 |
| | | (*c*) Explain how missing data were addressed | 7 |
| | | (*d*) If applicable, describe analytical methods taking account of sampling strategy | 6 |
| | | (*e*) Describe any sensitivity analyses | 6 |
| **Results** | | | |
| Participants | 13* | (a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed | 7 |
| | | (b) Give reasons for non-participation at each stage | 7 |
| | | (c) Consider use of a flow diagram | 25 |
| Descriptive data | 14* | (a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders | 7 |
| | | (b) Indicate number of participants with missing data for each variable of interest | 7 |
| Outcome data | 15* | Report numbers of outcome events or summary measures | 8 |
| Main results | 16 | (*a*) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included | 8 |

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| | | | |
|---|---|---|---|
| | | (*b*) Report category boundaries when continuous variables were categorized | 8 |
| | | (*c*) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period | 8 |
| Other analyses | 17 | Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses | 8 |
| **Discussion** | | | |
| Key results | 18 | Summarise key results with reference to study objectives | 9 |
| Limitations | 19 | Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias | 12 |
| Interpretation | 20 | Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence | 10 |
| Generalisability | 21 | Discuss the generalisability (external validity) of the study results | 12 |
| **Other information** | | | |
| Funding | 22 | Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based | 1 |

*Give information separately for exposed and unexposed groups.

**Note:** An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at http://www.plosmedicine.org/, Annals of Internal Medicine at http://www.annals.org/, and Epidemiology at http://www.epidem.com/). Information on the STROBE Initiative is available at www.strobe-statement.org.

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